

JIA WADHWA

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EDUCATION

University of Toronto

Toronto, ON

Bachelor of Applied Science in Computer Engineering, minor in AI and Engineering Business

2022 – Expected 2027

Courses: Operating Systems (Linux, Unix), Data Structures and Algorithms, Software Architecture, Computer Architecture, Fundamentals of Deep Learning, Probability and Statistics

Awards: University of Toronto International Scholar Award, UTEK Consultancy Competition (3rd place)

SKILLS

Cloud & DevOps: AWS (EC2, S3), GCP (Vertex AI, Firestore), Docker, Linux/Unix, Git/GitHub, Postgres, Unit Testing, Debugging

Programming & Data: Python, C++, C, JavaScript, HTML/CSS, SQL, MySQL, Oracle, pandas, NumPy

Software Engineering: System Design, Software Architecture, Object-Oriented Programming, Algorithms and Data Structures, SDLC

AI & Agentic Systems: LangGraph, LangSmith, MCP, Prompt Engineering, PyTorch, TensorFlow, Keras

PROFESSIONAL EXPERIENCE

Artificial Intelligence Intern | Airbus UK

Bristol, UK | Aug 2025 – Aug 2026

- Containerized microservices, the front-end and the back-end with Docker, achieving reproducible builds and streamlined cross-platform deployments.
- Deployed and managed applications on AWS EC2 instances and S3 buckets, ensuring scalability and efficient data handling.
- Designed and built an agentic AI framework with LangGraph and Google Cloud (Vertex AI, Firestore, Postgres) that explains RL agent decisions and surfaces trust metrics.
- Architected the database layer and hosted tools through MCP, cutting data retrieval time from 35s to 21s.
- Implemented MCP for secure tool calling within enterprise data-governance requirements.
- Used LangSmith to observe and evaluate agent responses; built a custom Python library to evaluate trustworthy-AI metrics for RL models.
- Fine-tuned and evaluated the Octo robot policy for a manufacturing use case, improving success rate from 60% to 80%.

Machine Learning Intern | iMax Technologies

New Delhi, India | Jul 2024 – Aug 2024

- Developed an AI chat model for an Enterprise Resource Planning (ERP) system so managers can retrieve information efficiently. Reorganized data storage and optimized Oracle SQL queries for faster access.
- Built a Sequential LSTM price-prediction model in Keras achieving under 10% error.

PROJECTS

Project Manager | Airbus Project Management Challenge

Bristol, UK | Dec 2025 – Jul 2026

- Led a cross-functional team of 5, evaluating technical, operational, and schedule feasibility while keeping 100% alignment with client requirements.
- Developed a proactive risk mitigation strategy to catch technical bottlenecks early and hit all interim milestones ahead of schedule.

Software Developer | LazyWay: GIS System

Toronto, ON | Jan 2024 – Apr 2024

- Coded in C++ by leveraging OSM (OpenStreetMap) database API to create a global mapping system.
- Optimized API performance, ensuring efficient data lookup and retrieval by using algorithms like A* and Dijkstra.
- Performed rigorous tests and debugging to achieve 42% faster retrieval speed than required.

CropShield: Plant Disease Detection System

Toronto, ON | Jun 2024 – Aug 2024

- Engineered an image recognition model in Python by utilizing deep learning frameworks like transformers, CNN's by using PyTorch and TensorFlow libraries.
- Conducted rigorous research on various machine learning models like YOLO, ResNet to choose the best model.
- Analyzed data and visualized results using matplotlib library to hypertune the model and achieve 93% accuracy.

Hospital Management System

New Delhi, India | Jan 2022

- Built a patient and staff records system in Python and MySQL (30% faster retrieval) with documentation for non-technical users.

Mastermind | Embedded Systems Project

Toronto, ON | Apr 2024

- Coded a game in embedded C and created digital logic on FPGA hardware and VGA display.
- Conducted error-checking and utilized UI skills to make the VGA display.
- Integrated audio functionalities and hardware components with the game logic showcasing proficiency in embedded systems to make an engaging user experience.